



Fakultät für Informatik
Lehrstuhl für Echtzeitsysteme und Robotik

Numerical Inverse Kinematics in Cluttered Environments

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Seminar *Cyber-Physical Systems* ST 2023

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Submission: 31. July 2023

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Abstract—Nowadays, robots are often modeled and built redundantly with multiple degrees of freedom (DOF) needing to move around in complex environments. This report presents an overview of the most common numerical approaches to the Inverse Kinematics (IK) problem in cluttered environments. The mentioned use cases are active research topics based on the IK problem, making it indispensable.

Depending on the task, the IK problem can result as unsolvable, in a single solution, or a multitude of solutions, preventing a general analytical approach. Herefore many numerical methods have been proposed, which we will analyze and compare in this report. Due to the multimodality of the problem, we will further restrict ourselves to the class of Jacobian methods, heuristic methods, and meta-heuristic methods. Each method class will be introduced and inspected for the secondary obstacle avoidance task. Afterward, the presented methods will be compared to each other.

Results show that Heuristic and Meta-Heuristic Methods are the best options for solving the IK problem numerically in cluttered environments in a general way.

I. INTRODUCTION

In various widely spread fields, the Inverse Kinematics (IK) problem is predominant. Let it be the control of figures and simulations in the animation industry, an agriculture robot in charge of harvesting fruits without hitting any branches, an assisting surgical robot that is supposed to move to a target position without hurting the patient, or many more. For simplicity, we will restrict ourselves to the field of robotics. Nonetheless, the presented methods may also be compatible with different applications using a modified version.

Ever since the industry's increasing robot automation [1], the IK of numerous manipulators have to be calculated. Manipulators are often built with many degrees of freedom (DOF) to solve various complex tasks. That ends in the IK of a kinematically-redundant manipulator varying from being unsolvable to having infinite solutions. Only some manipulators IK can be calculated analytically [2]. Although existing approaches exist to solve the IK analytically [3] [4], recent research trends have shown that numerical methods are not restricting with increasing DOF. Since analytical methods have to be calculated for each manipulator individually, high engineering fees arise each time a robot is adapted even slightly. In contrast, numerical methods have vast application possibilities since the modification needed for a manipulator is manageable and implemented way faster than solving the system analytically. Based on that, we will further restrict ourselves to numerical methods for solving the IK problem.

In general, we will be classifying the presented methods into three categories. First is the class of Jacobian methods [2], which uses the Jacobian matrix to solve the IK of a given manipulator. The most popular approaches include the Jacobian Inverse, the Jacobian Transpose, and the (selectively) Damped Least Square method. The second class of solutions is the class of heuristic methods. We will look into the cyclic coordinate descent method (CCD) first introduced in [5], and the Forward And Backward Reaching Inverse Kinematics (FABRIK) method [6]. Third, the meta-heuristic methods will be presented. Here we will have a look at the Differential Evolutionary Algorithms(DEAs) [7] and the Particle Swarm Optimization (PSO) method [8]. One has to note that alternative methods exist that will not be discussed, for example, the Newton Methods [9], the Monte Carlo Methods [10], Style or Mesh-based IK [11] [12], the Sequential Inverse Kinematics method [13] [14], and Feedback IK [15].

Furthermore, the previously mentioned kinematically-redundant manipulators must cope with more challenges when used in the real world [16]. That includes (but not exclusively) the optimization of calculation time, minimization of energy consumption [17], joint limit avoidance [18], and many more [19] [20]. One is obstacle avoidance in cluttered environments, which has been researched actively [21]. Two prominent examples are agriculture robots [21] and surgical robots [22].

The former is a result of the increasing demand for field cultivation. Especially seasonal crops are still in need of manual harvest. Despite recent advances in automation technology, the regional and seasonal workforce is coping with challenges like increasing salaries and hard working conditions for humans. Opposed to that, robots can harvest fruits individually, possibly around the clock without payment [23] [24] and without physically injuring anyone. In addition, the manipulator must reach the target while avoiding collisions with branches, trunks, or other obstacles. All in one, the literature discloses that robots have been used widely for harvesting applications [25] [26] [27].

Surgical Robots would enable highly specialized doctors to operate on patients in remote places [28] in preoperative planning [29]. The IK problem with obstacle avoidance can be found when minimizing contact with the patient's body [30]. Moreover, during surgery, it can warn the surgeon of any human error and improve the surgery results [31].

This report is structured as follows. Section 2 explains the inverse kinematics problem and obstacle avoidance strategies.

Section 3 introduces and analyzes numerical methods for solving the inverse kinematics problem and discusses how obstacle avoidance can be combined. Here the presented methods will be compared in terms of usability for different use cases. Section 5 concludes and notes further real-world restrictions to consider.

II. THE INVERSE KINEMATICS PROBLEM AND OBSTACLE AVOIDANCE

A. The Inverse Kinematics problem

The Inverse Kinematics (IK) problem is an essential aspect of robotics, aiming to receive a robot's or animated figure joint configurations corresponding to a desired target position. Unlike the forward kinematics problem, which involves calculating the position and orientation of the end-effector given the joint angles, the IK problem poses a more challenging task of finding the joint angles that produce a desired end-effector pose. Because a manipulator can possibly reach a certain position with various joint angles, the IK problem can vary from infinite solutions to unsolvable.

Solving the IK problem analytically can be complex and computationally expensive, especially for complex robotic systems with many DOF [32]. Even though solving the IK problem analytically is highly accurate, some manipulators might lack an analytical solution, especially redundant manipulators [33]. As a result, numerical approaches have gained significant attention due to their ability to handle complex kinematic structures and provide efficient solutions. Although some research exists for analytical solvers [3] [4], we will focus on current trends of the numerical approaches for solving the IK problem.

Numerical methods for solving the IK problem involve iterative algorithms that gradually refine a starting estimate of the joint configurations until the desired end-effector position is achieved. These approaches rely on numerical optimization techniques, which aim to minimize an objective function that quantifies the discrepancy between the expected and achieved poses. On top, further constraints are often applied to these functions to fulfill specific requirements, one of which is obstacle avoidance in a cluttered environment.

B. Obstacle avoidance

In many use cases, manipulators have to operate in cluttered environments. Therefore a safe and efficient operation has to be ensured, by further constraining the IK problem.

1) *Primary and Secondary Tasks*: To achieve obstacle avoidance, it is essential to consider the concept of primary and secondary tasks. Primary tasks refer to the primary objectives of the robotic system, such as reaching a desired end-effector pose. In contrast, secondary tasks encompass additional purposes, such as avoiding collisions with environmental obstacles or constraints from the robot's mechanical structure, e.g., joint limits.

While the user often specifies the primary task, secondary tasks usually constrain the execution to comply with further restrictions. The challenge lies in finding an efficient solution

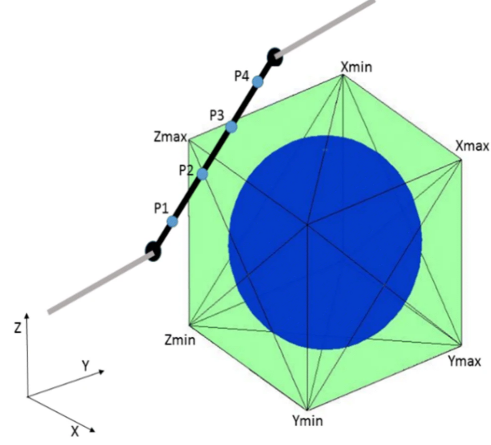


Fig. 1. from [34]. TFOBB of a spherical obstacle next to a manipulator link.

for the primary tasks that complies best with the secondary task. That also includes collision-free movement.

2) *What is Obstacle Avoidance*: Obstacle Avoidance ensures that the robotic system's movements to their end-effector pose are collision-free while conserving the constraints of the robot's mechanical structure. Although the IK problem usually deals with the end-effector position, collisions can occur on every robot link. The most common approach is utilizing the tight-fitting oriented bounding box method. In the following subsection, we will discuss this method in more detail.

3) *Tight-fitting Oriented Bounding Box*: The tight-fitting oriented bounding box (TFOBB) method is often used for obstacle avoidance in IK [34] for simplifying calculations. It involves constructing bounding boxes around the robot's links and the surrounding obstacles (Fig. 1). By comparing the bounding boxes, potential collisions can be detected. This can be helpful when we need to perform operations that may be inefficient to execute on the actual geometries due to the number of vertices. Especially when avoiding collision with humans, real-time calculations are needed in which the real geometries would be too time-consuming. The TFOBB method aims to find joint angles that maximize the distance between the robot's links and the obstacles, ensuring that the IK solving is collision-free. Moreover, the TFOBB method offers fast computation and is particularly suitable for environments with well-defined geometric obstacles.

However, it may be less effective in environments with irregular or complex shapes, where the tight-fitting bounding boxes may need to accurately represent the collision regions. Research shows that optimizations have attempted to solve that problem by using various geometric forms [35] [36].

TFOBB being a secondary task, we still need to explain methods for solving the IK problem as a primary task. This will be done in the following section.

III. NUMERICAL METHODS FOR SOLVING THE INVERSE KINEMATICS PROBLEM

The last section introduced the TFOBB method for obstacle avoidance that can be put on top of the existing method to

solve the IK problem. Many methods have been developed to refine the joint configurations to reach the end-effector pose. Based on a m -dimensional task space and n joint angles, we will define in a vector $\theta \in R^n$ for a manipulator with n DOF. The following will introduce an overview of the most common numerical approaches for solving the IK problem and how the secondary task of obstacle avoidance can be achieved using them. These include the Jacobian Methods, Heuristic Methods, and Meta-Heuristic Methods. We will not discuss other methods classes in this report. Exemplarily, not discussed methods are the Newton Methods [9] [10] which is an optimization approach, the Monte Carlo Method [10], the Sequential Monte Carlo Method [37], Neural Networks [38], the Style or Mesh-based Method [11] [12], the sequential Inverse Kinematics method [14], and Feedback IK [15].

A. Jacobian Methods

The Jacobian methods are the most commonly used class of solving techniques used for the inverse kinematics problem. These methods involve calculating the Jacobian matrix, representing the relationship between the joint and end-effector velocities. The relation between the end-effector's velocity $\dot{s}$ and the joint velocity $\dot{\theta}$ can be given by:

$$\dot{s} = J\dot{\theta} \quad (1)$$

where $J \in R^{m \times n}$ is the Jacobian matrix which can be seen as a function $J(\theta)$ of the angles θ . We search the change in joint angles $\Delta\theta$ that can be estimated by $\Delta s \approx J\Delta\theta$ when choosing $\Delta\theta$ in a way so that Δs is approximately equal to $\vec{e}$ by linearization. Then IK problem can be solved when the Jacobian matrix is symmetric by reconfiguring the equation (1) to:

$$\Delta\theta = J^{-1}\vec{e} \quad (2)$$

Otherwise, we can take advantage of the kinematically-redundant end-effectors by applying the following numerical Jacobian methods, each with strengths and limitations.

1) *Jacobian Transpose*: The Jacobian transpose method involves calculating the transpose of the Jacobian matrix and using it to find a solution for the change in joint angles. It was introduced by [39] [40] and calculates the change in joint angles using the following equation:

$$\Delta\theta = \alpha J^T \vec{e} \quad (3)$$

where α is a positive scalar, and base idea is to replace the inverse with the transpose of the Jacobian matrix. If α is small enough, [40] and [39] showed that the error of approximating the inverse with the transpose is neglectable by justifying the use in terms of virtual forces. According to [41], it has a lower computational burden than fellow Jacobian methods since no inverses need to be calculated. However, it may not always provide an optimal solution, especially when multiple end-effectors come into play.

2) *Jacobian Pseudo Inverse*: The Jacobian pseudo-inverse method, also known as the Moore-Penrose inverse, calculates a pseudo-inverse of the Jacobian matrix J because J is most likely redundant [42] and non-square. Thus, an ordinary inverse is impossible. The pseudo-inverse of the Jacobian matrix J^+ can be calculated as follows:

$$J^+ = J^T(JJ^T)^{-1} \quad (4)$$

The change in joint angles can then be calculated using the following equation:

$$\Delta\theta = J^+ \vec{e} \quad (5)$$

However, this method is volatile near singularities, which occurs when the Jacobian matrix is about to lose full rank. That usually happens when the end-effector is almost entirely extended, resulting in large changes in joint angles. Nonetheless, the Jacobian pseudo inverse method won't move in an impossible direction precisely at singularities and is, therefore, well-behaved.

Additionally, the matrix $(I - J^+J)$ projects onto the nullspace of J . Applied to robotics, the nullspace comprises all changes in joint angles that result in no changes at the end effector position or orientation. Using the following equation, [43] first used the property that $J(I - J^+J)\phi = 0$ to avoid joint limits:

$$\Delta\theta = J^+ \vec{e} + (I - J^+J)\phi \quad (6)$$

This is called the nullspace method, and ϕ can be used for secondary tasks and/or prioritizing various tasks [44] [45].

In practice, the pseudo inverse method is rarely used because of the near singularities' difficulties. Therefore, the Singular Value Decomposition and (selectively) Damped Least Squares methods are preferred due to their superior performance [46].

3) *Singular Value Decomposition*: Singular value decomposition (SVD) is a technique to decompose a matrix into constituent parts. In essence, the SVD assimilates the Pseudo Inverse Method rather than being a method in itself. It can be used to calculate the pseudo-inverse of the Jacobian matrix, which can then be used to solve the inverse kinematics problem. The SVD of the Jacobian matrix, J , can be calculated as follows:

$$J = U\Sigma V^T \quad (7)$$

where U and V are orthogonal matrices, and Σ is a diagonal matrix containing the singular values of J . The pseudo-inverse of J can then be calculated as follows:

$$J^+ = V\Sigma^+ U^T \quad (8)$$

where Σ^+ is the pseudo inverse of Σ . SVD is particularly useful when dealing with ill-conditioned or singular matrices because the singular value decomposition of the Jacobian always exists. This will be used to deduce the Selectively Damped Least Square method [46].

4) *Damped Least Squares*: The damped least squares (DLS) method involves adding a damping factor to the Jacobian matrix to improve the stability of the solution by preventing large joint velocities and avoiding many of the Pseudo Inverse method's problems. It was first introduced by [47] and is based on the formula:

$$\Delta\theta = J^T(JJ^T + \lambda^2 I)^{-1}\vec{e} \quad (9)$$

where λ is a positive damping factor, and I is the identity matrix. Adding I ensures the existence of an inverse, but λ has to be chosen carefully. Many methods have been proposed to determine λ ([48] [49] [50] [51] [52] [53] [54]). The equation will result in jerky movements and a slow convergence rate when λ is too small, but λ must also be large enough to guarantee that the solution is well-behaved near singularities. The advantages of the DLS method include that it still embodies the Pseudo Inverse Method when setting $\lambda \approx 0$ and when λ approaches infinity in the equation (9) the impact of the Jacobian term JJ^T diminishes and the equation approximates to

$$\Delta\theta = J^T(\lambda^2 I)^{-1}\vec{e} \quad (10)$$

The system is heavily damped in this configuration, with the change in joint changes scaled down by λ^2 . This can be useful in scenarios where the Jacobian is ill-conditioned, preventing big changes in joint angles that could result from numerical instability. However, the trade-off is a slower response to errors in the end effector's position.

TABLE I
TEST RESULTS WITH REACHABLE TARGET POSITIONS

Method	Wins	Mean Excess	Mean # iterations for accuracy of			
			0.1	0.01	0.001	0.0001
SDLS	2972	0.0	39.3	48.7	52.2	53.5
DLS	0	0.00015	61.5	231.6	467.1	635.2
J^T	0	2.362	44.0	51.6	53.4	53.9

Table adapted from [46]

TABLE II
TEST RESULTS WITH UNREACHABLE TARGET POSITIONS

Method	Wins	Mean Excess	Mean # iterations for accuracy of			
			0.1	0.01	0.001	0.0001
SDLS	940	0.182	173.5	336.0	548.1	686.0
DLS	52	0.106	116.1	448.9	1166.7	1706.2
J^T	0	4.907	44.0	142.27	142.30	142.30

Table adapted from [46]

5) *Selectively Damped Least Squares*: The selectively damped least squares (SDLS) method presented by [46] is a combination of the SVD and DLS methods. It allows for the selective damping of individual joints to improve the stability of the solution and prevent large joint velocities in specific joints. More into detail, the SDLS method decides, based on the distance from the end effector to the target position, how harsh the dumping factors will be for each joint. That reduces the jittery movements alongside singularities immensely. Those jittery movements are unimportant when

checking whether a solution for the IK exists. [46] stated the superior quality compared to the Transpose method (Table I & II) because the Transpose method started to oscillate near singularities comparatively early. On top of that, its position precision is superior to the DLS and Transpose methods' precision.

Conversely, when low precision is expected, the DLS method converges in fewer iterations. It is therefore suggested to first use the DLS method and switch to the SDLS method when the target position is nearly reached.

6) *Obstacle Avoidance with Jacobian methods*: Regarding Jacobian methods, research shows two techniques to extend the IK problem with obstacle avoidance [55]. On the one hand, there are the Null-Space based methods [21] [56]. On the other hand, the obstacle avoidance task can be implemented as an optimization problem with an objective function [57] [58] [59]. Although the Null-space methods are comparatively faster, they suffer from the singularity problem. Based on that, optimization techniques are recommended for better quality results. These will not be considered in this report but among the widely used are Flexible priority solution [57], weighted least-norm method [60], gradient projection method [43], Sequential Quadratic programming [61] and the closed-loop inverse kinematics algorithm [43]. Mathematically the objective function of the IK problem while avoiding collision with the environment can be given as the following minimization function:

$$H = \left\| J\dot{\theta} - \dot{\vec{s}} \right\|^2 + \left\| \rho\dot{\theta} \right\|^2 + \left\| J_c\dot{\theta} - \dot{\vec{s}}_c \right\|^2 \quad (11)$$

where the first term is the difference between the desired end effector position deducible from $\vec{s}$ and the actual position deducible from $J\dot{\theta}$ (see III-A). This describes the IK problem solved using the Jacobian Inverse. The second term was added to avoid singularity problems, where $\dot{\theta}$ is calculated using the DLS method weighted with ρ . Alternatively, one could replace the two terms using one of the others methods mentioned above. The third term incentivizes the robot doesn't collide with the environment. $J_c \in R^{c \times c}$ is the Jacobian matrix for obstacle avoidance, and c represents the number of links that could possibly collide with obstacles. More closely, this term is based on the TFOBB method, where the distance between the manipulator and the created box of the obstacle is investigated.

[34] explains that the potential collision point between the i -th joint and the center of the obstacle's bounding box can be given as:

$$d_{ci} = \|s_{ci} - s_0\| \quad (12)$$

where s_0 is the coordinate of the obstacle in the task space and s_{ci} is the link point that could collide and is defined as $s_{ci} = s_i + p_i e_i$, where s_i is the coordinate of the i -th joint. p_i is the projection of the line from the i -th joint of the manipulator link to the center of a particular obstacle on the i -th link expressed as $p_i = e_i^T(s_0 - s_i)$. Further, that can be used to determine the potential point of collision to the center of the obstacle:

$$u_i = \frac{s_{ai} - s_0}{d_{ci}} \quad (13)$$

The Jacobian matrix J_c can then be formed with the submatrices J_{ci} . J_{ci} is expressed for each row i and the corresponding angle q :

$$J_{ci} = -u_i^T \frac{\delta s_{ci}}{\delta q} \quad (14)$$

One algorithm was proposed by Gottschalk et al. [62]. All in one, the IK problem as a primary task and obstacle avoidance as a secondary task are active topics in research.

It should be noted that this delivers a local solution and does not guarantee global optimality. Furthermore, we shall consider that there are many optimization possibilities and techniques [63] [64] [57] which are outside the scope of this report. While numerous methods are available, choosing the right one depends on the application's requirements. Especially when looking at the first two terms which describe the IK problem, the following can be deducted:

TABLE III
COMPARISON OF JACOBIAN METHODS FOR IK PROBLEM

Method	Complexity	Singularity Robustness	Precision
Pseudo Inverse	Medium	Low	Low
Transpose	Low*	Low	Low
DLS	Medium	Medium	Medium
SDLS	High	High	High
DLS + SDLS	Medium	High	High

* Only with one end-effector. For multiple end-effectors, other methods are superior.

As depicted in Table III, the Transpose method should be chosen if real-time performance is most important. Although it has not the equal singularity robustness and precision compared to the other ways, it has the lowest computational complexity [65]. Introducing multiple end-effectors reveals a superior DLS and/or SDLS runtime compared to the Transpose method. When precision is the priority, and a certain runtime can be accepted, SDLS is recommended. A combination of DLS and SDLS could be used to reduce that runtime. The DLS method can be preferred to the SDLS method when a faster implementation and runtime is required. However, the SDLS method converges the fastest when the target position can not be reached. The Pseudo Inverse method has its advantages based on the low computational complexity but is prone to inferior quality due to the mentioned singularity problem and is therefore rarely used.

B. Heuristic Methods

Heuristic methods have emerged as a pragmatic alternative to traditional Jacobian numerical solutions for solving the IK problem. These methods employ rule-based or experience-based strategies to explore the search space of potential solutions, leveraging computational efficiency and the flexibility to handle complex, high-degree-of-freedom systems. While exact solutions may not always be guaranteed due to the inherent approximations, these heuristic methods often provide sufficiently accurate and feasible solutions for many practical robotics applications.

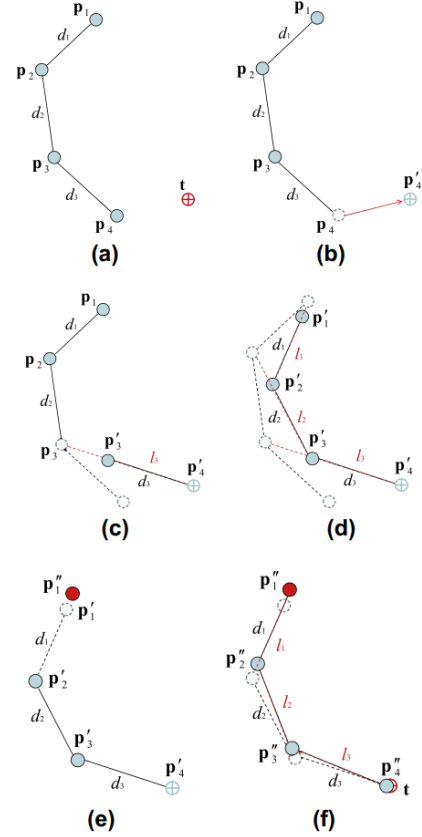


Fig. 2. from [6]. An example of a full iteration of FABRIK for the case of a single target and 4 manipulator joints. (a) The initial position of the manipulator and the target, (b) move the end effector p_4 to the target, (c) find the joint p'_3 which lies on the line l_3 that passes through the points p'_4 and p_3 , and has distance d_3 from the joint p'_4 , (d) continue the algorithm for the rest of the joints, (e) the second stage of the algorithm: move the root joint p'_1 to its initial position, (f) repeat the same procedure but this time start from the base and move outwards to the end effector. The algorithm is repeated until the position of the end effector reaches the target or gets sufficiently close.

1) *Cyclic Coordinate Descent*: Cyclic Coordinate Descent (CCD) is a heuristic method [5] [66] widely used for solving inverse kinematics problems in robotics as well as in the computer graphics industry. It minimizes a joint error at a time while simultaneously holding all others constant, cycling through the joints until a satisfactory solution with a given error is achieved.

Based on its nature, CCD can result in various configurations. Therefore, restricting constraints have to be implemented to receive a feasible configuration. Despite those constraints, CCD can produce unnatural movements and suffers from motion with erratic discontinuities. CCD was initially designed for serial chains and has difficulties with multiple end-effectors. [67] proposed a possible workaround. Since each joint computation is low, the solution can be exerted with a swift runtime. On top CCD has a simple code implementation, a linear-time complexity in DOF, and is numerically stable.

2) *FABRIK*: Forward And Backward Reaching Inverse Kinematics (FABRIK) [6] is a heuristic method that iteratively

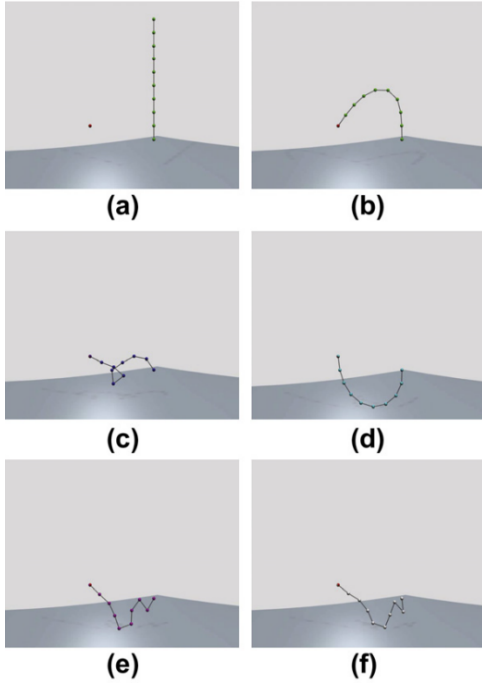


Fig. 3. from [6]. Experimental solutions using some of the most popular IK methods. The kinematic chains consisted of 10 unconstrained joints, allowing 3 DOF on each joint. (a) Initial position, (b) FABRIK, (c) CCD, (d) Transpose, (e) DLS, (f) SDLS

adjusts the joint angles in both forward and backward directions until the end-effector reaches the desired position (Fig.2) with an acceptable error margin.

The bidirectional approach in FABRIK, and the iterative adjustments, make it suitable for solving multiple end-effector problems. Additionally, FABRIK reduces the potential for erratic motion (Fig. 3) compared to CCD by treating the kinematic chain as a whole in each iteration. FABRIK also exhibits linear time complexity in the number of iterations and the DOF. The FABRIK method is noted for its superior computational efficiency and speed [6].

It is a popular choice for real-time applications in robotics, game development, and the animation industry [6]. The iterative nature of FABRIK allows it to adaptively adjust the joint angles based on the current state of the kinematic chain, thus generating more natural and smooth motions (Fig. 3). In spite of this, the quality of solutions still relies on the fine-tuning of parameters within the algorithm. Looking at an increasing pose precision of the end-effector in the dimension of $10^{-4}m$, FABRIK reveals an unstable convergence behavior [68].

3) *Obstacle Avoidance with Heuristic methods:* Heuristic methods often incorporate strategies for avoiding obstacles while still reaching the target position. Like their Jacobian-based counterparts, heuristic methods have been extended for obstacle avoidance in inverse kinematics. This report will focus on the FABRIK method since it surpasses CCD in complexity, movement quality, and speed [6]. FABRIK also does not contain the constraints problem, is compatible with multiple end effectors, and is superior in all comparable

aspects [6]. FABRIK has been modified to incorporate obstacle avoidance capabilities in [69] [70] and for continuum robots [71] and [72] also introduced an extension. Still, we will concentrate on non-continuum robots in this report.

A two-step obstacle avoidance technique was proposed by [69] to extend the basic FABRIK method. Firstly, a heuristic approach is adopted to ensure the updated linkage bar, the root joint, and the target position in a line so that interference can be eliminated in most cases. Secondly, multiple random rotation strategies are applied to eliminate the interference that has not been eliminated in the first step.

However, the extended FABRIK method still needs to address the problem that the experimental results may vary for different tests due to adopting to avoid the obstacle. There are also some optimization approaches [73] [74] that further improved FABRIK while maintaining the secondary task obstacle avoidance.

4) *Comparison of Heuristic and Jacobian methods:* Due to the IK's comparably low computation time a FABRIK surpasses all Jacobian methods by far in runtime [74]. Adding obstacle avoidance, we can also assume a faster calculation using the FABRIK method compared to the Jacobian method when looking at the given time of 0.99s to 1.51s for a 5-link manipulator in [57] and 0.07s to 0.2s for a 4-link manipulator in [69]. In conclusion, using the FABRIK approach, we can assume more natural movements (Fig.3) and faster calculation times. On top heuristic methods are free from singularities methods and therefore recommended when compared to any Jacobian method considering obstacle avoidance.

C. Meta-Heuristic Methods

We will define the class of Meta-Heuristic Methods for any Heuristic approach not explicitly designed for the IK problem. These methods have shown promising results, mainly due to their global search capability and the ability to handle multi-objective and multimodal optimization problems. Meta-heuristic methods offer a stochastic global optimization approach that will be discussed in the following section, often involving nature-inspired strategies. Their defining characteristic is their reliance on a general minimization function, which aids in their application across various complex and disparate problem domains. The function guides the search through the solution space to find an optimal or near-optimal solution, effectively minimizing the objective function, which in the case of inverse kinematics, would be to reduce the error between the desired and actual end-effector position.

$$\min (||P_{cur} - P_{des}||^2 + ||R_{cur} - R_{des}||^2) \quad (15)$$

where P_{cur} is the current manipulator position, and P_{des} is the target position of the end-effector. The subtraction is the Euclidean distance between P_{cur} and P_{des} . Analogously, R_{cur} is the current orientation of the manipulator, and R_{des} is the target orientation. Optionally those terms can be weighted as used in [75]. Methods that will not be discussed include Simulated Annealing [76], Genetic Algorithm [77], the Ant

Colony Optimization [78] [79], Bees Algorithm [80] [81] [82], and optimization methods mentioned in III-A6.

1) *Differential Evolution Algorithms*: Differential Evolution Algorithms (DEAs) [83] are a class of population-based stochastic algorithms for global optimization. The key idea behind DEAs is the generation of new candidate solutions by combining existing ones. Specifically, DEAs produce offspring by adding the weighted difference of two randomly selected solutions to a third one. The offspring are then evaluated and compete with the current generation's individuals for a place in the next generation:

- 1) Initialization: A population of potential solutions is randomly initialized.
- 2) Mutation: This stage generates a 'donor' vector by adding the weighted difference of two randomly chosen vectors to a third one.
- 3) Crossover: This process combines the donor vector with a target vector to create a 'trial' vector.
- 4) Selection: If the trial vector's fitness is greater than or equal to the target vector's fitness, the trial vector replaces the target vector in the next generation.

A crucial aspect of DEAs is the control parameters: mutation scaling factor, crossover rate, and population size. The mutation scaling factor influences the rate at which the population evolves. The crossover rate affects the diversity of the offspring, while the population size determines the algorithm's ability to explore the solution space. Compared to Evolutionary Algorithms, DEAs specifically use vector difference operations and a simple arithmetic crossover to manipulate the population of solutions. DEAs have proven effective in solving the IK problem [7] [84] due to their ability to handle non-linear, non-differentiable, multi-modal, and high-dimensional spaces. Despite their advantages, DEAs also have limitations, such as being sensitive to parameter settings and slow convergence in the later stages of optimization. That can be mitigated through adaptive or self-adaptive strategies dynamically adjusting the control parameters based on the algorithm's performance.

2) *Particle Swarm Optimization*: Particle Swarm Optimization (PSO) [85] is a meta-heuristic optimization technique that draws its inspiration from the social behavior of bird flocks or fish schools. PSO operates by maintaining a swarm of particles that traverse the search space. Each particle in the swarm represents a potential solution to the problem at hand. The position of a particle is updated based on its personal best solution and the global best solution found by the swarm so far. PSO is noted for its simplicity, ease of implementation, and the small number of parameters required. It has shown significant effectiveness in solving the IK problems [8] [86]. However, like most meta-heuristics, PSO can sometimes struggle with premature convergence and lack of diversity in the swarm. Various strategies, such as adaptive inertia weight, dynamic parameter adaptation, and hybridization with other algorithms, have been proposed to address these limitations [8].

3) *Obstacle Avoidance with Meta-Heuristic methods*: Similarly to the minimizing function for the Jacobian methods,

the object function can easily be extended to avoid collision. The new formula can be expressed as follows:

$$\min \left(\|P_{cur} - P_{des}\|^2 + \|R_{cur} - R_{des}\|^2 + \sum_{i=1}^m C_i \right) \quad (16)$$

where C_i is the penalty of the i -th collision and m is the amount of collisions. When comparing the presented method, the initial ascertainment may be that their performance is similar. [87] showed that combining DEAs and PSO will further optimize runtime and accuracy. Due to the versatile nature of the algorithms, the combination of PSO and DEAs [88] is also recommended for the IK problem considering obstacle avoidance

TABLE IV
COMPARISON OF JACOBIAN, HEURISTIC, AND META-HEURISTIC METHODS

Method	Runtime	Singularity Robust	Scalability
Jacobian	Medium	No	Low
Heuristic	Low	Yes	Medium
Meta-Heuristic	High	Yes	High

4) *Comparing Meta-Heuristic methods with Jacobian Methods and Heuristic methods*: As seen in Table IV Meta-Heuristic methods do not have difficulties around singularities. However, based on the broad application possibilities, the runtime of the Meta-Heuristic methods is higher than the Jacobian methods. Being specifically designed for the IK problem, the Jacobian methods are more efficient. Thus, considering the scalability and implementation time, the Meta-Heuristic shine due to its adaptive nature. Further, it can be said that the Meta-Heuristic methods are more likely to avoid local minima due to their highly adapting nature and may therefore achieve a better target position while also coping with multiple end-effectors.

Comparing the Meta-Heuristic and Heuristic methods, we note that both methods are well-behaved around singularities. In contrast, the computational complexity is lower within the Heuristic methods since they were also designed for the IK Problem and prove a better efficiency. Even with this, the Meta-Heuristic methods may also result in a more precise target position when considering that Heuristic methods, especially with obstacle avoidance, may not find the optimal solution. In terms of scalability and ease of implementation, the Meta-Heuristic methods have the advantage compared to the Heuristic methods, which are not generally applicable.

In general, the Meta-Heuristic's robustness, easy implementation, and scalability make the Meta-Heuristic an excellent method candidate when the runtime is not important. Otherwise, the heuristic method can be recommended when prioritizing real-time usability.

IV. CONCLUSION AND FURTHER RESTRICTIONS

In this report, we discussed the most common Jacobian, Heuristic, and Meta-Heuristic Methods and analyzed which may be the best option for different situations. Concluding the Meta-Heuristic method, e.g. a combination of PSO and DEAs,

is recommended when the real-time calculation is not of the essence. If fast runtimes are essential, a Heuristic method, like an optimized form of FABRIK, is the best option. On top of that optimization techniques (III-A6,III-C4) can be combined with those two methods, further reducing calculation time or the number of iterations.

When using those methods on robots in practice, additional factors must be considered. Some further restrictions concerning the inverse kinematic model's secondary task include optimizing execution time, Minimizing energy consumption, actuation limits, contact dynamics, obstacle detection, path planning, joint limits, or dynamics of a robot, and many more.

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